according to their molecular weight. Each spot on the resulting 2D gel correspond to single protein species present in the sample (Fig.10.11)

Applications : SDS–PAGE can be used to determine the molecular weight of proteins. To achieve this, a standard mixture of proteins of various molecular weight (molecular weight ladder) was added for direct comparison of migration distance. The molecular weights around 15-200kDa can be analyzed in this manner. Another application of SDS PAGE is to check the purity of a protein sample. Presence of a single band denote the protein sample is pure.

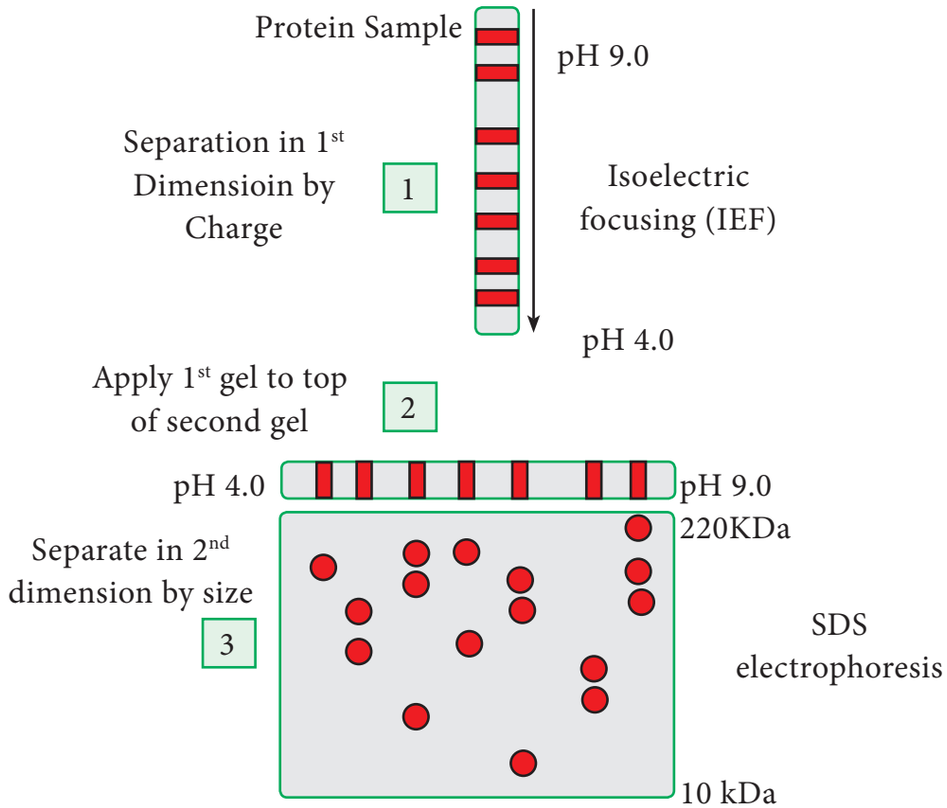

Fig.10.11 Two dimensional gel electrophoresis

iv Agarose gel electrophoresis

Agarose is one of the several components that can be separated from agar. The major source of agar is certain species of sea weed. Agarose is a linear polymer made up of alternating units of galactose and 3,6- anhydrogalactose. Agarose gels are completely transparent when cooled to room temperature.

In Agarose gel electrophoresis, DNA or RNA molecules can be separated based on their size. This is achieved by the movement of negatively charged nucleic acid molecules through an agarose matrix in a horizontal electrophoresis. Molecules with smaller size

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